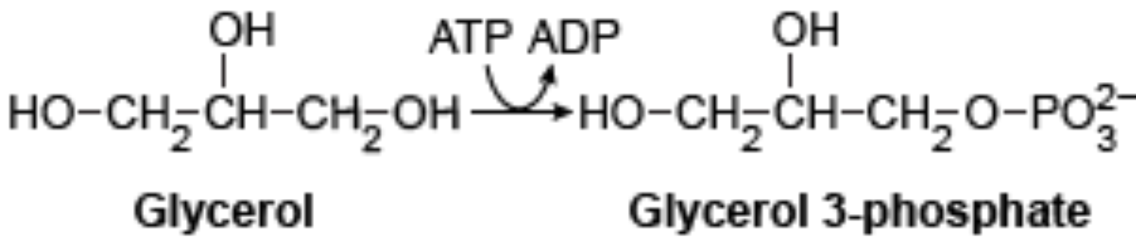
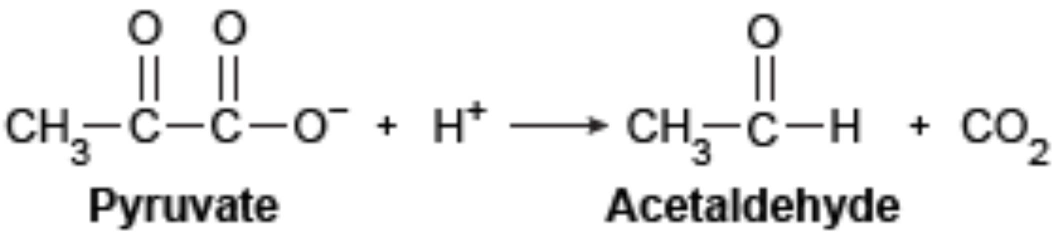
Fatty acids must be activated by a ligase enzyme prior to degradation. Which of the following reactions is catalyzed by a ligase?

A.



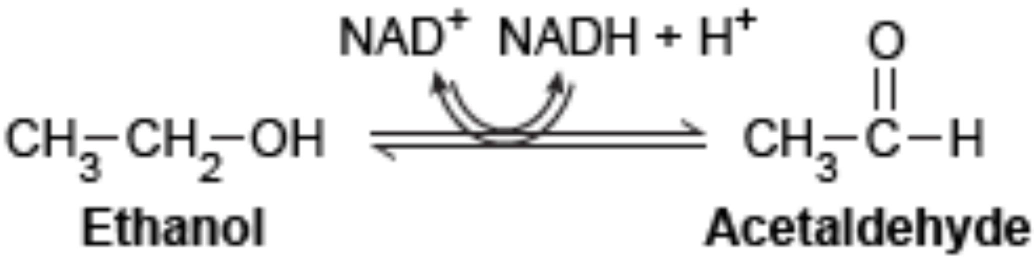
[18%]

B.



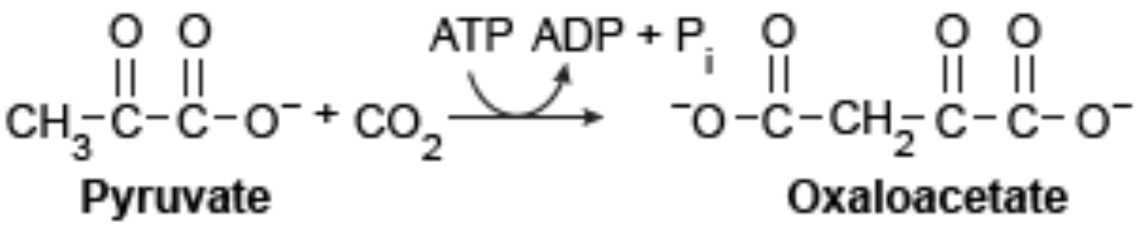
[12%]

C.



[3%]

D.



[65%]

Correct
65% answered correctly

Explanation:

Class	Reaction type	Mechanism
Oxidoreductases	Oxidation-reduction	$A^- + B \rightleftharpoons A + B^-$
Transferases	Functional group transfer	$A-B + C \rightleftharpoons A + B-C$
Hydrolases	Hydrolytic cleavage of a molecule into two molecules	$A-B + H_2O \rightleftharpoons A-H + B-OH$
Lyases	Group removal or addition with electron rearrangement to form double bonds	$\overset{X}{\underset{}{A}}-\overset{Y}{\underset{}{B}} \rightleftharpoons A=B + X-Y$
Isomerases	Functional group movement within a molecule	$\overset{X}{\underset{}{A}}-\overset{Y}{\underset{}{B}} \longrightarrow \overset{Y}{\underset{}{A}}-\overset{X}{\underset{}{B}}$
Ligases	Joining two molecules to form one via bond formation and ATP hydrolysis	$ATP + A + B \rightleftharpoons A-B + ADP + P_i$

The **ligase** class of enzymes catalyzes addition or synthesis reactions between two molecules by coupling them with the **hydrolysis of energy-rich bonds** such as those between **phosphate groups of nucleotides like ATP**. Pyruvate carboxylase catalyzes the energy-requiring carboxylation of pyruvate to oxaloacetate, an intermediate in amino acid synthesis, gluconeogenesis, and the citric acid cycle. Because pyruvate carboxylase uses the energy of ATP to "join" pyruvate and CO_2 to form oxaloacetate, the enzyme is classified as a ligase.

(Choice A) In this reaction, the transferase glycerol kinase moves a phosphate group from ATP to glycerol to form glycerol 3-phosphate. Transferases are used to shift functional groups from one molecule to another.

(Choice B) One CO_2 molecule is removed from pyruvate by the lyase pyruvate decarboxylase. Lyases catalyze the cleavage of single molecules into two molecules (via breakage of C–C, C–O, C–N, and C–S bonds). These enzymes do not require water or energy from ATP, and can form double bonds in their products.

(Choice C) Alcohol dehydrogenase, an oxidoreductase, catalyzes the reduction of acetaldehyde to ethanol via concomitant oxidation of NADH to NAD^+. Oxidoreductases catalyze reactions by facilitating the transfer of electrons between molecules with cofactors such as NAD^+, $NADP^+$, and FAD^+.

Educational objective:
The ligase class of enzymes catalyzes addition or synthesis reactions between two molecules by coupling them with the hydrolysis of energy-rich bonds such as those between ATP phosphate groups.

Time spent:0 sec
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